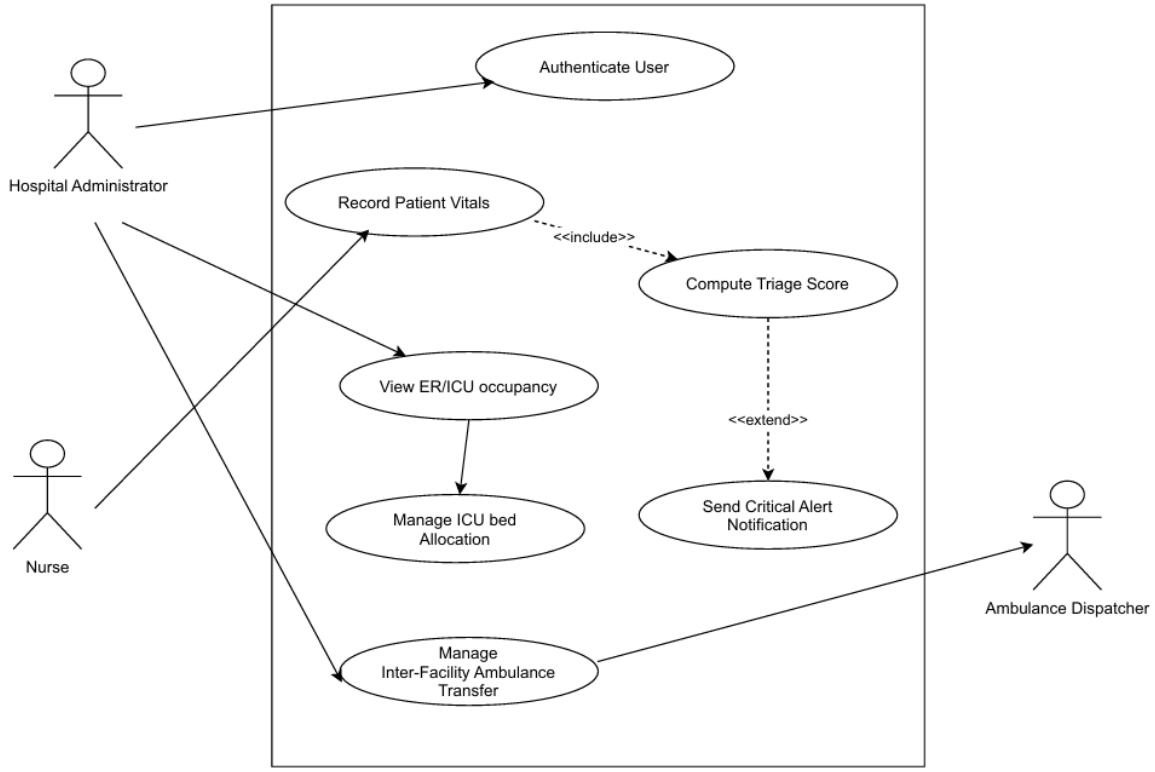
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Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.	High	Pass: Higher acuity patient automatically moves ahead of lower acuity patient. Fail: FIFO queue ignores critical emergency.	Core safety-critical function; ensures the sickest patients are treated first rather than in arrival order.
FR-002	Functional	The system shall allow an authenticated Triage Nurse to record patient intake vitals (heart rate, blood pressure, SpO2, temperature, GCS) through a structured entry form.	High	Pass: All six vital fields are validated and saved with a timestamp within 5 seconds of submission. Fail: An incomplete or out-of-range vitals record is accepted without a validation error.	Accurate, structured vitals capture is the required input for triage score computation (FR-001).
FR-003	Functional	The system shall display real-time ward and ICU bed occupancy status (Occupied, Cleaning, Available) to Hospital Administrators on a live dashboard.	High	Pass: A bed status change made at any client is reflected on every connected dashboard. Fail: The dashboard shows a bed status that is more than 1 second out of date.	Administrators need an accurate, real-time view of capacity to make fast bed-allocation decisions.
FR-004	Functional	The system shall allow a Hospital Administrator to assign an available ICU bed to the highest-ranked waitlisted patient and mark that bed as Occupied.	High	Pass: Assigning a bed removes the patient from the waitlist and immediately marks the bed Occupied for all users. Fail: Two administrators are able to assign the same bed to two different patients at the same time.	Prevents double-booking of a scarce, safety-critical ICU resource during high-pressure situations.
FR-005	Functional	The system shall allow a Hospital Administrator to initiate and track an inter-facility ambulance transfer request, capturing the receiving facility, dispatch time, and current transfer status.	Medium	Pass: A submitted transfer request produces a trackable status (Requested / En-route / Completed) visible on the dashboard and to the Ambulance Dispatcher. Fail: A transfer request exists with no visible status.	Supports a safe, auditable handoff when local ICU capacity is unavailable and the patient must be moved.

NFR-001	Nonfunctional -Performance & Security	Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected ER dashboard clients in under 1 second via WebSockets.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Stale bed-status information directly risks patient safety by delaying ICU transfers during emergencies.
NFR-002	Nonfunctional – Security	The system shall restrict access to patient vitals, triage scores, and bed-allocation functions to authenticated users, enforcing role-based permissions for Triage Nurse and Hospital Administrator roles.	High	Pass: Unauthenticated requests and out-of-role actions (e.g., a Triage Nurse assigning an ICU bed) are denied and logged. Fail: A user can view or modify data outside their assigned role.	Protects sensitive patient health information and prevents unauthorized bed reassignment, in line with healthcare data-protection obligations.

UML Diagram



Use-Case

Main Flow

1. Triage Nurse selects “Record Patient Vitals” and enters intake vitals for the patient.
2. System validates the entered vitals for completeness and plausible ranges.
3. System computes a triage severity score from the validated vitals.
4. System inserts the patient into the ER bed allocation waitlist, ordered by descending severity score.
5. System updates the ER dashboard in real time so all connected clients see the new waitlist order.
6. Hospital Administrator reviews the waitlist and current ICU/ward occupancy on the dashboard
7. Hospital Administrator selects the highest-ranked waitlisted patient and assigns an available ICU bed
8. System marks the selected bed as Occupied, removes the patient from the waitlist, and propagates the change to all dashboard clients within 1 second.
9. Use case ends successfully.

Alternate Flow- 3a: Critical Triage Score

- System determines the computed triage score exceeds the critical-acuity threshold.
- System immediately triggers UC-07 to the on-duty Hospital Administrator and relevant clinical staff, in parallel with waitlist ranking.
- Hospital Administrator acknowledges the alert and expedites ICU bed assignment for the patient.
- Flow resumes at main scenario step 7.

Alternate Flow-7a: No ICU Bed Available

- Hospital Administrator finds no ward/ICU bed with status Available for the patient's required bed type.
- Hospital Administrator initiates an inter-facility ambulance transfer request, specifying the receiving facility.
- System records the transfer request status as “Requested” and notifies the Ambulance Dispatcher.
- Patient remains on the ER waitlist, flagged as “Pending Transfer,” until the ambulance transfer is completed or a local bed becomes available.